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PERI-OPERATIVE

Peri-Operative Management of Diabetes

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Key Points

- See JBDS-IP March 2021 guidance for more information

General Principles

- Diabetic patients should be prioritised on the theatre list.
- It is crucial to assess how long the patient is expected to be fasting.
- Minor surgery: total period of starvation missing one meal.
- Major surgery: total period of starvation missing two or more meals.

Principles of treatment

- Before surgery, if the HBA_{1C} >69mmol/mol, then refer to Diabetes Specialist Team for optimisation. If out of hours, then start a Variable Rate Intravenous Insulin Infusion (VRIII).
- Around the day of surgery follow the instructions outlined in the tables 1 & 2 below.
- Patients with planned short starvation period (one to two missed meals) should be managed by modification of their usual medications, avoiding VRIIIs where possible.
- Use clinical discretion when considering starting a VRIII. VRIII is preferable before surgery if any of the following applies:
  - Blood glucose is ≥12mmol/L.
  - Expected starvation period of more than one meal missed.
  - Emergency surgery is planned.
  - Have a low threshold for setting up a VRIII for patients with HbA_{1c} >69mmol/mol.
- When indicated, insert iv cannula and start VRIII with intravenous fluid containing potassium (unless serum K^+ >5.0mmol/L) at least one hour before surgery.
- Check capillary blood glucose (CBG) level every hour (two hours if on non-insulin therapy).

Glucose Control

- CBG levels need monitoring hourly on the day of the surgery and whilst on VRIII. This can be reduced to every four hours when patient is stable and back on subcutaneous insulin therapy.
- Maintain blood glucose levels between 6-12 mmol/L.
- If CBG is >12mmol/L:
  1. Exclude Diabetic Ketoacidosis (capillary ketones >3mmol/L) & Hyperosmolar Hyperglycaemic State (osmolality >320mOsm/kg)
  2. Administer 2 to 6 units (0.1units/kg if Type 2 diabetes) subcutaneous rapid analogue insulin (Novarapid). **Assume that one unit will drop CBG by 3mmol/l.**
  3. Repeat CBG (& ketones, if elevated) one hour later. If CBG is still >12mmol/l, consider starting VRIII.
- Hypoglycaemia (CBG<4mmol/L; Looming Hypoglycaemia: 4-6mmol/L):

- Give 15-20g glucose orally, if conscious, orientated & able to swallow, otherwise, IV

Electrolyte Control

- Serum electrolytes should be measured daily whilst patient on sliding scale insulin.
- Fluids must be prescribed in accordance with the protocol for intravenous sliding scale insulin.

Discontinuation of VRIII

- When patient is eating again, convert from intravenous to subcutaneous (s/c) insulin at time of meal.
- Continue sliding scale insulin for approx 30-60 minutes after administering s/c insulin, then stop.
- DO NOT STOP VRIII WITHOUT RESTARTING PATIENTS' USUAL S/C INSULIN.
- Oral Antidiabetic agents can be restarted as soon as patient can tolerate solid foods. *(unless oral antidiabetic agent is contra-indicated)*

Intravenous fluids and sliding scale protocol for diabetic surgical in-patients

Intravenous fluids

Prescribe Glucose 5%/Sodium Chloride 0.45% with Potassium depending on the plasma Potassium levels. *(see table below)*

Plasma K^+ (mmol/L)	Dose
≤3.5	40mmol/L (0.3%)
3.6-5.0	20mmol/L (0.15%)
>5	No added potassium

Connections

- Connect insulin with intravenous fluids using a flexible Y set connector with one way and anti-siphon valves.
- Do not use 3-way taps or separate cannulae.
- It is important to flush the end of the Y connectors with the appropriate fluid.

Patients transfer *(e.g. between departments)*

- Once the regime has started it is best to use it continuously.
- If intravenous fluid is stopped for any reason the insulin needs to be stopped as well.

VRIII Preparation

Prepare human soluble insulin, 50 units in 50ml sodium chloride 0.9% in a 50mL syringe.

(*When available please use pre filled syringes from pharmacy)*

Infuse using a syringe driver at a rate according to the sliding scale prescribed on treatment sheet.

The insulin **preparation is only stable for 12hours**, after which the patient should be reviewed and a new syringe of insulin prepared if necessary.

Patient Observations

Please follow the all Wales **'Hourly insulin check chart'** available on your wards.

Table 1- Management of subcutaneous insulin therapy during a starvation period

*If the patient requires an ongoing VRIII, then the long acting background insulin should be continued, but reduce usual dose by 20%. Normal insulin dose should be recommenced when the patient is eating and drinking normally.

Insulins	Day prior to surgery	Day of Surgery IF NOT ON VRIII		Day of Surgery If patient is on a VRIII*
		Patient for morning surgery	Patient for afternoon Surgery	
Once Daily (evening) (e.g. Lantus® or Levimir®, insultard®, Humulin I®, Insuman®)	Reduce usual evening dose by 20%.	Check blood glucose on admission	Check blood glucose on admission Resume usual insulin dose if patient eating and drinking as normal.	Reduce usual dose by 20%.*
Once Daily (morning) (e.g. Lantus® or Levimir®, Insultard®, Humulin I®, Insuman®)	No dose change required	Reduce usual dose by 20%. Check blood glucose on admission	Reduce usual dose by 20%. Check blood glucose on admission	Reduce usual dose by 20%.*
Twice daily (e.g. Novomix 30®, Humulin M3®, Humalog Mix 25®, Humalog Mix 50®, Insuman Comb 25®, Insuman Comb 50®)	No dose change required	Halve the usual morning dose. Check blood glucose on admission. Resume normal insulin regime with the evening meal.	Halve the usual morning dose. Check blood glucose on admission. Resume normal insulin regime with the evening meal	Stop until eating and drinking normally
Twice daily – Separate injections of short acting (e.g. Animeal neutral, NovoRapid® Humulin S® Aprida®) and intermediate acting (e.g. animal isophane Isultard® Humulin I® Insuman®)	No dose change required	Calculate the total dose of both morning insulins and give half as intermediate acting only in the morning. Check blood glucose on admission Leave the evening meal dose unchanged	Calculate the total dose of both morning insulins and give half as intermediate acting only in the morning. Check blood glucose on admission Leave the evening meal dose unchanged	Stop until eating and drinking normally
3,4 or 5 injections daily (e,g. An injection of mixed insulin 3 times a day or 3 meal time injection of short acting insulin and once or twice daily background)	No dose change required	Basal Bolus regimens: Omit the morning and lunchtime short acting insulins. If the dose of long acting basal insulin is usually taken in the morning, then dose should be reduced by 20% Premixed am insulin: halve the morning dose and omit lunchtime dose Check blood glucose on admission	Take usual morning insulin dose(s) Omit lunchtime dose Check blood glucose on admission	Stop until eating and drinking normally
Short acting and rapid acting insulin (e.g. Novorapid®, Humulin S®, Apidra®)	No dose change required	Omit while the patient nil by mouth. Restart usual regime when patient eating and drinking as normal.	Take usual morning dose with breakfast. Omit while the patient nil by mouth. Restart usual regime when patient eating and drinking as normal.	Stop until eating and drinking normally
Continuous subcutaneous insulin infusion pumps.	No dose change required	Infusion pump should be removed and VRIII initiated. Close liaison with the diabetic team is recommended.		

Table 2- Management of non-insulin medication during a starvation period

Medication	Day prior to Surgery	Day of Surgery IF NOT ON VRIII		Day of Surgery If a VRIII is being used
		Patient for morning surgery	Patient for afternoon Surgery	
Acarbose	Take as normal	Omit morning of the operation if nil by mouth	Give morning dose if eating.	Stop once VRII commenced, do not recommence until eating and drinking normally.
Meglitinde (repaglinide or neteglinide)	Take as normal	Omit morning of the operation if nil by mouth	Give morning dose if eating.	
Metformin	Take as normal	Omit on day of surgery	Omit on day of surgery	
Metformin M/R	Omit evening before surgery	Omit on day of surgery	Omit on day of surgery	
Sulphonylurea (e.g. Glibenclamide, Gliclazide, Glipizide, etc)	Take as normal	Omit the morning dose	Omit on day of surgery	
Gliclazide M/R	Omit evening before surgery	Omit on day of surgery	Omit on day of surgery	
Pioglitazone	Take as normal	Take as normal	Take as normal	
DPP IV inhibitor (e.g. Sitagliptin, Vidagliptin, Saxagliptin, alogliptin, linagliptin)	Take as normal	Take as normal	Take as normal	
GLP-1 analouge (e.g Exenatide, Liraglutide, lixisenatide, dulaglutide)	Take as normal	Take as normal	Take as normal	Take as normal
SGLT2 inhibitor (e.g.canagliflozin,dapagliflozin, empagliflozin)	take as normal	Omit on day of surgery	Omit on day of surgery	Omit until eating and drinking normally

References

Joint British Diabetes Societies: Perioperative Care for People with Diabetes Mellitus Undergoing Elective and Emergency Surgery, March 2021